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Deploying Search and Chat Apps using Agent Builder

Course · 4 hours · < 1% complete

Course overview

Challenge Labs

Build and deploy a Website Search Application: Challenge Lab

Deploy a Chatbot using AI Applications: Challenge Lab

Create a Custom Search App using the Discovery API: Challenge Lab

Your Next Steps

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Course Survey

Course · Deploying Search and Chat Apps using Agent Builder ·

Quick tip: Review the prerequisites before you run the lab

End Lab00:49:24

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

Open Google Cloud Console

Username

student-02-e481aa0b23e1f

Password

HQ4cFZTejgx5

GCP Project ID

qwiklabs-gcp-01-c2d88e3f

Build and deploy a Website Search Application: Challenge Lab

★★★★☆ Rate Lab

This lab may incorporate AI tools to support your learning.

Lab instructions and tasks100/100

Challenge Lab Overview

Objective

Setup and requirements

Challenge Scenario

Task 1. Create an AI Applications Data Store

Task 2. Create and configure a Search App

Task 3. Deploy Search Widget to Public Webpage

Congratulations!

Challenge Lab Overview

or following step-by-step instructions, you are given a common business scenario and a set of tasks - you figure out how to complete them on your own! An automated scoring system (shown on this page) provides feedback on whether you have completed your tasks correctly.

When you take a Challenge Lab, you will not be taught Google Cloud concepts. You will need to use your skills to assess how to build the solution to the challenge presented. This lab is only recommended for students who have those skills. Are you up for the challenge?

Objective

This assessment will test your ability to create and deploy Search applications using AI Applications.

Setup and requirements

For each lab, you get a new Google Cloud project and set of resources for a fixed time at no cost.

1. Make sure you signed into Qwiklabs using an **incognito window**.
2. Note the lab's access time (for example, **02:00:00** and make sure you can finish in that time block.

There is no pause feature. You can restart if needed, but you have to start at the beginning.

3. When ready, click **START LAB**.

Open Google Console

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

Console.

Username

google2876526_student@qwiklabs.n

Password

TG959yrKDX

GCP Project ID

qwiklabs-gcp-0855e773352d3560

New to labs? View our introductory video!

5. Click **Open Google Console**.

6. Click **Use another account** and copy/paste credentials for **this** lab into the

If you use other credentials, you'll get errors or **incur charges**.

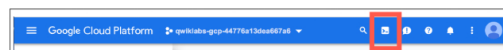
7. Accept the terms and skip the recovery resource page.

Do not click **End Lab** unless you are finished with the lab or want to restart it. This clears your work and removes the project.

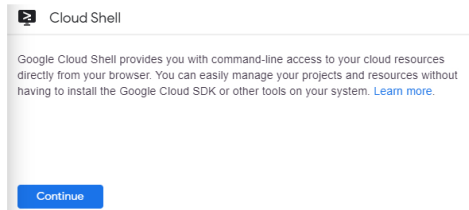
Activate Cloud Shell

Cloud Shell is a virtual machine that is loaded with development tools. It offers a persistent 5GB home directory and runs on the Google Cloud. Cloud Shell provides command-line access to your Google Cloud resources.

In the Cloud Console, in the top right toolbar, click the **Activate Cloud Shell** button.



Click **Continue**.



It takes a few moments to provision and connect to the environment. When you are connected, you are already authenticated, and the project is set to your *PROJECT_ID*. For example:



Shell and supports tab-completion.

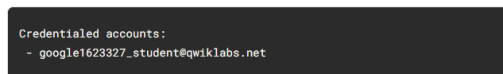
You can list the active account name with this command:



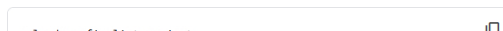
(Output)



(Example output)



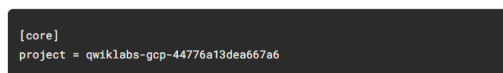
You can list the project ID with this command:



(Output)



(Example output)



For full documentation of `gcloud` see the [gcloud command-line tool overview](#).

Challenge Scenario



Here is a company overview as provided on the Cymbal Shops's website.

Cymbal Shops is an American retail chain headquartered in Minneapolis that sells homeware, electronics, and clothing.

Founded in 1974, Cymbal Shops started out as Cymbal Air, selling AC systems manufactured by Cymbal Group in Minnesota and neighboring states. The company quickly expanded into domestic merchandise in order to satisfy the need for quality products at an affordable price in the midst of a recession.

Cymbal Shops' broad product assortment was once a benefit - capturing planned purchases and last minute splurges. In recent years, however, the company has struggled to adapt to the acceleration in e-commerce. Digitally native companies are on the rise and Cymbal Shops must implement significant changes in order to keep pace and maintain relevance. Today, Cymbal Shops operates 714 stores

Cymbal Shops is a digitally transforming legacy retailer.

It is inspired by clients like: Bed Bath & Beyond, Best Buy, Home Depot, Nordstrom.

Your challenge

Cymbal Shops is looking for proficiency in creating and deploying search applications using AI Applications and Conversational Agents. You work with Cymbal Shops, and are tasked with developing a custom search app utilizing the Dialogflow API to interact with a Vertex AI Data Store and present search results to users.

Key technologies and Google Cloud services to be utilized include:

- AI Applications
- Search Apps
- Agents
- Gemini

Upon successful completion of this challenge, you will demonstrate proficiency in:

- Deploying Search Applications.

Task 1. Create an AI Applications Data Store

In this section, you will utilize the AI Applications to create a Data Store. Data stores act as knowledge bases, storing information from websites and documents that your agent can use to answer user questions. Therefore in this task,

1. Enable [AI Applications API](#).
2. Create a data store that will be used by a generative agent to answer user questions. Configure the data store using the following details:

Property	Value
Data Store Name	cymbal-data-store

Location	global (Global)
----------	-----------------

Click **Check my progress** to verify the objectives.

Create an AI Applications Data Store

Check my progress

Assessment Completed!

Task 2. Create and configure a Search App

In the previous section, you created a data store filled with valuable resources on Python programming. Now, let's leverage that knowledge base to build a custom search application specifically for programmers!

Imagine the possibilities - programmers can quickly find solutions, troubleshoot problems, and explore new concepts, all within a user-friendly search interface.

Therefore in this task:

- 1. Create an application of type Search with the following configurations. This application will use the data store named cymbal-data-store.

Property	Value
App Type	Search for your website
Enterprise edition features	Selected
Advanced LLM features	Selected
Name the app	cymbal-search-app
Company Name	cymbal-python-programming-company
Location	global (Global)
Data Store	cymbal-data-store

- 2. Preview your app and ask it a question similar to: What is the syntax of a

Note: It may take a short time for the Data Store to be available. If you get an error, wait a couple of minutes and try again.

Output:

What is the syntax of a Loop in Python?

https://www.educba.com/python/python_for_range...

Python Looping Through a Range

To loop through a set of code a specified number of times, we can use the range() function. The range() function returns a sequence of numbers, starting from 0 ...

https://www.educba.com/python/for_break_app

Python break Keyword

Example: Get your own Python Server. End the loop if it is larger than 3. for i in range(0): if i > 3: break print(i) Example: Break out of a while loop: i = 1

https://www.educba.com/python/python_dictionary_loop...

Python - Loop Dictionaries

What is a correct syntax for looping through the values of this dictionary x = {'cyril': 'frank'}

https://www.educba.com/python/python_if_statement...

Python If Statement

These conditions can be used in several ways, most commonly in 'if statements' and loops. An 'if statement' is written by using the if keyword.

https://www.educba.com/python/python_syntax_error...

Invalid Syntax in Python: Common Reasons for SyntaxError - Real ...

The messages "break outside loop" and "continue not properly in loop" help you figure out exactly what to do. If this code were in a file, then Python would ...

https://www.educba.com/python/python_for_while_app

Click Check my progress to verify the objectives.

✓

Create and configure a Search App

Check my progress

Assessment Completed!

Task 3. Deploy Search Widget to Public Webpage

In this section, we'll focus on deploying the Search Widget hosted on a web page. This will make the application accessible online.

- 1. In the Google Cloud Console, navigate to AI Applications > Integration. To deploy the search widget, refer to the code snippet of the web application.

the allowed domain. Copy the code highlighted for your Web Application and save your changes.

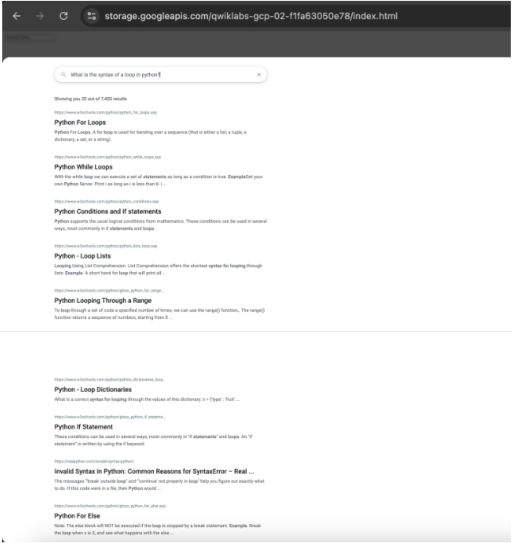
- 3. Use Cloud Shell to create a new file named index.html. Copy and paste the generated code into the index.html file.
- 4. Create a Cloud Storage bucket named qwiklabs-gcp-01-c2d88e379459

Property	Value
Bucket Name	quwiklabs-gcp-01-c2d88e379459
Location	Multi-region(us)
Storage Class	Standard
Access Control	Uniform

5. Grant the `Storage Object viewer` role to `allusers` on the `qwiklabs-gcp-01-c2d88e379459` bucket.
6. Upload the `index.html` file in your Google Cloud Storage bucket using the `gsutil` command.

`c2d88e379459/index.html`.

Note: It may take a few minutes for the allowed domain to be available. Please try again if you encounter an error.



Click **Check my progress** to verify the objectives.

Deploy a Search Widget to Public Webpage

Check my progress

Assessment Completed!

Congratulations!

Congratulations! You've successfully completed the challenge lab, adeptly demonstrating your ability to use AI Applications to deploy search applications.

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Lab Last Tested December 12, 2024

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